E

Engineering Insights

Edition #1 — 2026-05

TL;DR

- Established Forward Deployed Academy to train engineers in AI deployment, enhancing client alignment and operational goals.
- Implemented selective AI tools in Software Development Lifecycle to improve code quality and productivity, with ongoing training.
- Adopted cloud first strategy for enterprise architecture, enhancing scalability and innovation while aligning with business outcomes.
- Embraced vibe coding with AI tools, enhancing productivity and flexibility in software development, establishing standard procedures.
- Initiated comprehensive AI training and metrics to boost productivity, achieving significant time savings and process improvements.

Integrating Forward Deployed Engineers with AI: Bridging the Gap

In our journey to enhance the integration of AI systems within enterprise environments, we encountered a significant challenge: bridging the gap between complex AI technologies and the specific business needs of our clients. This is where the concept of Forward Deployed Engineers (FDEs) became crucial. Unlike traditional engineers who focus on internal product development, FDEs are technical professionals who work directly on-site with clients to customize, deploy, and operationalize complex software or AI systems. Their role is pivotal in ensuring that AI solutions are not only implemented but also aligned with the client's operational goals and business strategies [c1].

As we explored the potential of FDEs, we realized the importance of training our tech leads to become proficient in this role. This involved discussions on the necessary skills and knowledge required, such as understanding AI tools, cloud deployment, and containerization technologies like Docker. We also considered the need for role-specific training, ensuring that project managers and business analysts could effectively utilize AI in their respective domains. The goal was to create a comprehensive training program that would equip our team with the skills needed to effectively deploy AI solutions in real-world environments [c2].

The outcome of these discussions was the establishment of the FD Academy, which offers two distinct tracks: a technical track and a consulting track. This initiative aims to provide targeted training that addresses both the technical and business aspects of AI deployment. By doing so, we are not only enhancing our team's capabilities but also ensuring that we can deliver AI solutions that meet the specific needs of our clients. Moving forward, we plan to continue refining our training programs and exploring new ways to integrate AI into our service offerings, thereby maintaining our competitive edge in the rapidly evolving tech landscape [c3].

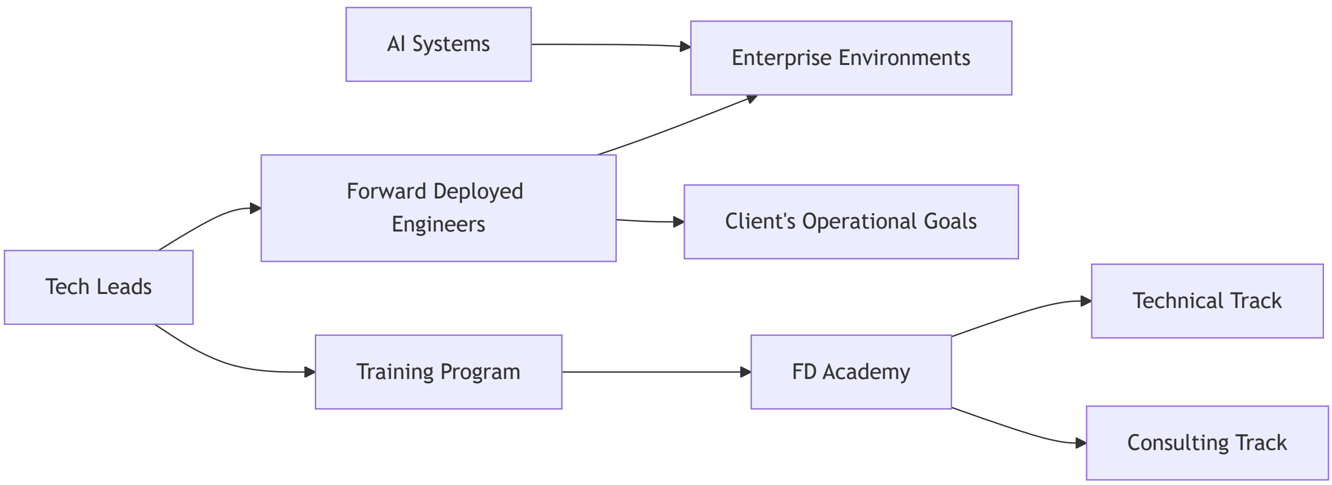

Diagram illustrating the integration of Forward Deployed Engineers with AI systems and the establishment of the FD Academy for training.

Integrating AI into the Software Development Lifecycle: Challenges and Opportunities

In recent years, the integration of Artificial Intelligence (AI) into the Software Development Lifecycle (SDLC) has become a focal point for many organizations seeking to enhance productivity and efficiency. The challenge we faced was how to effectively incorporate AI tools into our existing development processes to maximize their potential benefits. This was crucial because, as the demand for faster and more reliable software solutions grows, leveraging AI could provide a significant competitive advantage by reducing development time and improving code quality [c1].

During our exploration, we discussed various aspects of AI integration, including the use of AI for both new and existing modules. It was observed that AI tools tend to perform better when developing new modules from scratch, as they can better understand and adapt to the project structure and design patterns [c2]. However, when modifying existing modules, the results were less accurate, necessitating manual adjustments based on AI outputs. This highlighted the importance of understanding the limitations and strengths of AI tools in different phases of development [c3]. Additionally, we considered the role of AI in project management and how it could be tailored to different functions within the organization, such as business analysis and project management [c4].

Ultimately, we decided to implement AI tools selectively, focusing on areas where they could provide the most value, such as generating code variations and improving code quality through automated suggestions [c5]. We also recognized the need for ongoing training and adaptation, ensuring that all team members, not just developers, are equipped to leverage AI effectively in their roles [c6]. Moving forward, we plan to establish metrics to measure the success of AI integration, such as productivity gains and quality improvements, to continuously refine our approach [c7]. By doing so, we aim to not only enhance our current processes but also set a foundation for future innovations in AI-driven software development [c8].

Embracing a Cloud First Approach in Modern Enterprise Architecture

In recent years, our team has faced the challenge of modernizing enterprise architecture to keep pace with rapidly evolving technological landscapes. The traditional on-premises infrastructure, while reliable, often fell short in terms of scalability, flexibility, and cost-effectiveness. This was particularly evident as our projects grew in complexity and required more agile and innovative solutions to meet business demands. The need for a strategic shift became apparent, leading us to explore the cloud first approach, which prioritizes cloud solutions as the default option while allowing for exceptions when justified [c1].

As we delved into the cloud first strategy, we considered several factors. A key aspect was understanding that not all projects are suited for the cloud; some require on-premises solutions due to regulatory, performance, or integration needs [c2]. We also explored the technical benefits of cloud services, such as serverless computing and Platform as a Service (PaaS), which reduce operational overhead and improve scalability [c3]. The cloud first approach also aligns with a zero trust security model, enhancing security by granting access based on identity context and least privileges [c4]. Additionally, financial operations (FinOps) practices were crucial in providing cost visibility and optimization, preventing overspending while enabling innovation [c5].

Ultimately, we decided to adopt the cloud first approach, recognizing its potential to transform our enterprise architecture. This decision was supported by the benefits of improved scalability, faster time to market, and the ability to leverage cutting-edge technologies like artificial intelligence (AI) [c6]. As we move forward, our focus will be on continuous learning and collaboration across teams, fostering a culture that embraces change and innovation. By aligning our technology with business outcomes, we aim to drive growth, resilience, and competitiveness in the market [c7]. Our next steps involve creating a comprehensive framework to guide future projects, ensuring that our cloud first strategy is implemented effectively and consistently across the organization [c8].

Vibe Coding and AI Tools in Development: A New Era of Software Engineering

In recent months, our team has been exploring the burgeoning field of vibe coding, a methodology that has rapidly gained traction among enterprise customers and is now being embraced by smaller clients as well [c1]. The core of vibe coding lies in its flexibility and the use of AI tools to enhance productivity and code quality. This approach allows developers to act more as reviewers, leveraging AI agents to handle the bulk of coding tasks [c2]. The absence of strict tool restrictions or frameworks in vibe coding means that developers can choose the tools that best fit their needs, while still adhering to privacy policies [c3]. This flexibility is crucial as it allows for a more readable and optimized code output, which is a significant departure from traditional coding practices [c4].

As we delved deeper into vibe coding, we considered various AI tools that could support this new approach. Google AI Studio emerged as a promising option, offering capabilities to generate base code for applications through prompts and providing a gallery of applications to draw from [c5]. However, the challenge lies in ensuring that these AI-generated outputs are accurate and align with existing modules, especially when integrating new features into established systems [c6]. We also recognized the importance of maintaining a consistent theme in UI and UX design, which requires careful prompt engineering to prevent unnecessary deviations [c7].

Ultimately, our exploration led us to a consensus that AI tools significantly enhance productivity by offering multiple coding options and allowing developers to choose the most feasible solutions for their projects [c8]. This not only speeds up the development process but also broadens the developers' knowledge base by exposing them to various plugins and libraries [c9]. As we move forward, we aim to establish standard operating procedures (SOPs) to streamline the vibe coding process, covering everything from environment analysis to deployment [c10]. This structured approach will ensure that we maximize the benefits of AI tools in our software development lifecycle, paving the way for faster time-to-market and improved efficiency [c11].

AI Tools and Training for Enhanced Productivity

In our ongoing quest to enhance productivity within our software development lifecycle (SDLC), the integration of AI tools has emerged as a promising frontier. The question we faced was whether these tools could realistically boost our productivity by a factor of five. This ambitious target prompted us to explore the potential of AI not just in coding but across various stages of development and project management. The challenge was significant, as achieving such a leap in productivity would require not only the adoption of new technologies but also a cultural shift in how we approach our work [c1].

Our exploration began with a broad discussion on the potential of AI tools to transform our workflows. We considered the benefits of AI in generating multiple coding options, which allows developers to choose the most feasible solutions for their projects. This adaptability is crucial as it enables us to replace less efficient libraries with more suitable alternatives, thereby optimizing our codebase [c2]. However, the effectiveness of AI tools can vary depending on the task. For instance, they tend to perform better when developing new modules from scratch rather than modifying existing ones, as they can better understand and adapt to new project structures [c3].

Training and trust emerged as critical factors in leveraging AI tools effectively. We recognized the need to extend AI training beyond developers to include all roles within the organization, ensuring that everyone can harness AI's potential in their respective functions [c4]. Moreover, maintaining a structured approach to using AI, such as setting clear prompts and configurations, was identified as essential to avoid unnecessary outputs and maximize efficiency [c5].

The outcome of our discussions has been a strategic push towards comprehensive AI training and the establishment of metrics to measure productivity gains. By tracking metrics such as time saved and customer approval rates, we have already observed significant improvements in our processes. For example, in one project, we saved 197 hours through the use of AI, demonstrating its potential to streamline our operations [c6]. Moving forward, our focus will be on refining these metrics and expanding our AI training programs to ensure that all team members can contribute to and benefit from these productivity enhancements [C:4, C:12].

In conclusion, while the journey towards achieving a fivefold increase in productivity is complex, the integration of AI tools presents a viable path forward. By fostering a culture of continuous learning and adaptation, we are well-positioned to harness the full potential of AI, ultimately transforming our development processes and achieving our ambitious productivity goals.

Action Items

- **Recruitment Manager**: Inform RMG that Sachin is interested and capable of working with AWS.
- **Training Coordinator**: Develop a training program for tech leads to become Forward Deployed Engineers (FDEs), including prerequisites like AI, cloud platforms, and Python.
- **Solution Architect**: Explore the creation of an FD Academy with two tracks: technical and consulting, and share the framework with the team.
- **Technical Lead**: Ensure familiarity with cloud services, AI development tools, and Python for automation frameworks in DevOps and cloud computing.

- **Project Manager**: Identify senior candidates with experience in application development and customer problem-solving for FDE roles.

References

Integrating Forward Deployed Engineers with AI: Bridging the Gap

- [c1] `23056240-7777-4b6a-b01f-7cbae286e753` — Architects Innovation Day (4).docx {"timestamp": "11:01", "speaker_role": "Team Member"}
- [c2] `afa7d440-ca19-4c3e-a1d1-0a6e2113a256` — Architects Innovation Day (4).docx {"timestamp": "10:07", "speaker_role": "Team Member"}
- [c3] `8b40015c-ceea-4e65-a388-6cc40cf486f2` — Architects Innovation Day (4).docx {"timestamp": "22:11", "speaker_role": "Team Member"}

Integrating AI into the Software Development Lifecycle: Challenges and Opportunities

- [c1] `ca6269a3-62ad-46ba-93cd-2047b5803047` — Architects Innovation Day (4).docx {"timestamp": "20:32", "speaker_role": "Team Member"}
- [c2] `4403e388-56f2-4e6a-b35d-849d3cde3198` — Tech-Office - All Hands Meeting.docx {"timestamp": "13:19", "speaker_role": "Team Member"}
- [c3] `b9d7956d-3701-4041-9e4d-6dc4713b7049` — Tech-Office - All Hands Meeting.docx {"timestamp": "13:19", "speaker_role": "Team Member"}
- [c4] `ebd0d525-4177-4000-8a1b-3686eb8336aa` — Architects Innovation Day (7).docx {"timestamp": "25:06", "speaker_role": "Team Member"}
- [c5] `319469de-c8cd-4afd-b367-61f7f945acfb` — Tech-Office - All Hands Meeting.docx {"timestamp": "17:33", "speaker_role": "Team Member"}
- [c6] `d6f060a6-4808-47cd-8070-8f4ed46882bf` — Architects Innovation Day (7).docx {"timestamp": "26:36", "speaker_role": "Team Member"}
- [c7] `eaa749c7-31cf-462e-9743-db7a8203bde7` — Architects Innovation Day (7).docx {"timestamp": "32:29", "speaker_role": "Team Member"}
- [c8] `9a687d70-9a5f-43ba-818f-8ed2fc316a95` — Tech-Office - All Hands Meeting.docx {"timestamp": "0:03", "speaker_role": "Team Member"}

Embracing a Cloud First Approach in Modern Enterprise Architecture

- [c1] `0b6d1144-5a35-4ad8-92a2-9e905f947e8d` — Tech-Office - All Hands Meeting(2).docx {"timestamp": "1:40", "speaker_role": "Team Member"}
- [c2] `cd1d5e91-ec74-446e-991d-4f7acbbe61c7` — Tech-Office - All Hands Meeting(2).docx {"timestamp": "5:52", "speaker_role": "Team Member"}

- [c3] `3efab643-162e-4efa-abfe-e9fb1219e57a` — Tech-Office - All Hands Meeting(2).docx {"timestamp": "1:40", "speaker_role": "Team Member"}
- [c4] `b92ccec2-bcca-4a91-880c-2b6bfe22e6d0` — Tech-Office - All Hands Meeting(2).docx {"timestamp": "11:09", "speaker_role": "Team Member"}
- [c5] `08830152-8e4c-4c00-8c9e-8fbbf819deab` — Tech-Office - All Hands Meeting(2).docx {"timestamp": "11:32", "speaker_role": "Team Member"}
- [c6] `99000dca-840d-45fc-bcfb-6b1bd11ba17f` — Tech-Office - All Hands Meeting(2).docx {"timestamp": "1:40", "speaker_role": "Team Member"}
- [c7] `ec30b647-8f25-4219-abd3-40652f76b682` — Tech-Office - All Hands Meeting(2).docx {"timestamp": "11:32", "speaker_role": "Team Member"}
- [c8] `eeb9a623-2fcd-43e3-8325-5eb75beb7894` — Tech-Office - All Hands Meeting(2).docx {"timestamp": "16:17", "speaker_role": "Team Member"}

Vibe Coding and AI Tools in Development: A New Era of Software Engineering

- [c1] `abcfb0a5-3ec4-4351-aa60-385c64035de8` — Tech-Office - All Hands Meeting(2).docx {"timestamp": "37:39", "speaker_role": "Team Member"}
- [c2] `1926af65-f3b6-45b8-b001-3a304de3504b` — Tech-Office - All Hands Meeting(2).docx {"timestamp": "21:31", "speaker_role": "Team Member"}
- [c3] `daba9c12-857a-4a4d-a6a7-3f24eb516d99` — Tech-Office - All Hands Meeting(2).docx {"timestamp": "42:01", "speaker_role": "Team Member"}
- [c4] `e676ae36-13c3-42da-8063-c37cb3758d67` — Tech-Office - All Hands Meeting(2).docx {"timestamp": "22:31", "speaker_role": "Team Member"}
- [c5] `bbfc50d0-a213-4545-9fad-a5d52d98d496` — Tech-Office - All Hands Meeting(2).docx {"timestamp": "31:31", "speaker_role": "Team Member"}
- [c6] `109de2f4-a6b9-429a-92ad-8eab5c99aeaa` — Tech-Office - All Hands Meeting.docx {"timestamp": "13:19", "speaker_role": "Team Member"}
- [c7] `a52b18ee-b00c-4d9c-a680-51b5a11625fb` — Tech-Office - All Hands Meeting.docx {"timestamp": "20:24", "speaker_role": "Team Member"}
- [c8] `1bb20226-0d14-494d-a2bd-708d1ca0afad` — Tech-Office - All Hands Meeting.docx {"timestamp": "17:33", "speaker_role": "Team Member"}
- [c9] `35fe4bbd-01c0-4b70-aca7-a9cceb2ff8a4` — Tech-Office - All Hands Meeting.docx {"timestamp": "17:33", "speaker_role": "Team Member"}
- [c10] `0ebf526c-7d75-4f0b-9051-e80c354190f3` — Architects Innovation Day (7).docx {"timestamp": "27:35", "speaker_role": "Team Member"}
- [c11] `d4ac6359-4040-4af3-a18d-aaa6474eca7d` — Tech-Office - All Hands Meeting(2).docx {"timestamp": "27:09", "speaker_role": "Team Member"}

AI Tools and Training for Enhanced Productivity

- [c1] `0f1efc0d-a6cb-4726-89db-44a388ee82d2` — Tech-Office - All Hands Meeting.docx {"timestamp": "0:03", "speaker_role": "Team Member"}
- [c2] `dc8498db-55f8-4752-b085-24e14de65fec` — Tech-Office - All Hands Meeting.docx {"timestamp": "17:33", "speaker_role": "Team Member"}
- [c3] `bae25bcc-87b1-4af6-ab04-fa44de9f6efa` — Tech-Office - All Hands Meeting.docx {"timestamp": "13:19", "speaker_role": "Team Member"}
- [c4] `f717fcd1-8f78-49b2-ab0a-5ef8e5d0c4b3` — Architects Innovation Day (7).docx {"timestamp": "26:36", "speaker_role": "Team Member"}
- [c5] `a5a1af14-730c-461d-8bc9-4ee3b99ba225` — Tech-Office - All Hands Meeting.docx {"timestamp": "20:24", "speaker_role": "Team Member"}
- [c6] `02d27a2c-d158-42f7-823e-1e16007bec70` — Tech-Office - All Hands Meeting(1).docx {"timestamp": "57:21", "speaker_role": "Team Member"}
